UNTRM AI WORKSHOP · SESSION 4

Seeing with AI: From Raw Images to a Deployable Model

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Chachapoyas · Tuesday, November 10, 2026

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This session's plan

PART 1 · 15 MIN

The task and the data

Nematodes under the microscope

PART 2 · 20 MIN

Pre-label with SAM 2

A foundation model drafts the labels

PART 3 · 20 MIN

Re-annotate in Labelbox

Correct, don't draw from scratch

PART 4 · 25 MIN

Semi-supervised training

Pseudo-labels and retraining

PART 5 · 10 MIN

Evaluate

Model counts vs. expert counts

PART 6 · 20 MIN

Deploy with Streamlit

From a model file to an app

MOTIVATION · NEMA-AI

Nematodes are still counted by hand, under a microscope

40,000+
samples a year at the NCDA&CS Nematode Assay Lab

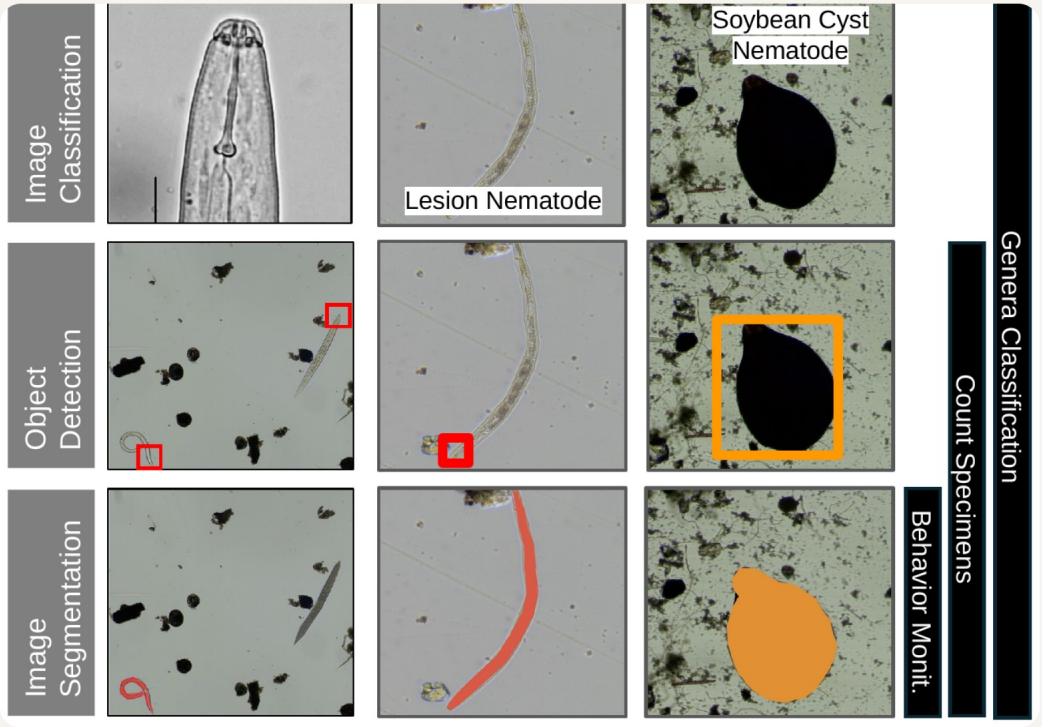
60–70
samples per technician per day, by eye

3–4
months growers can wait for results at peak season

Manual counting needs years of training, is tiring and error-prone, and delays decisions on rotation and nematicides.

NEMA-AI · IMAGE ANALYSIS TASKS

Classify, detect, segment, then count



What labs need

Which genus, how many, and how they behave

What studies report

Above 90% on most metrics, on clean samples

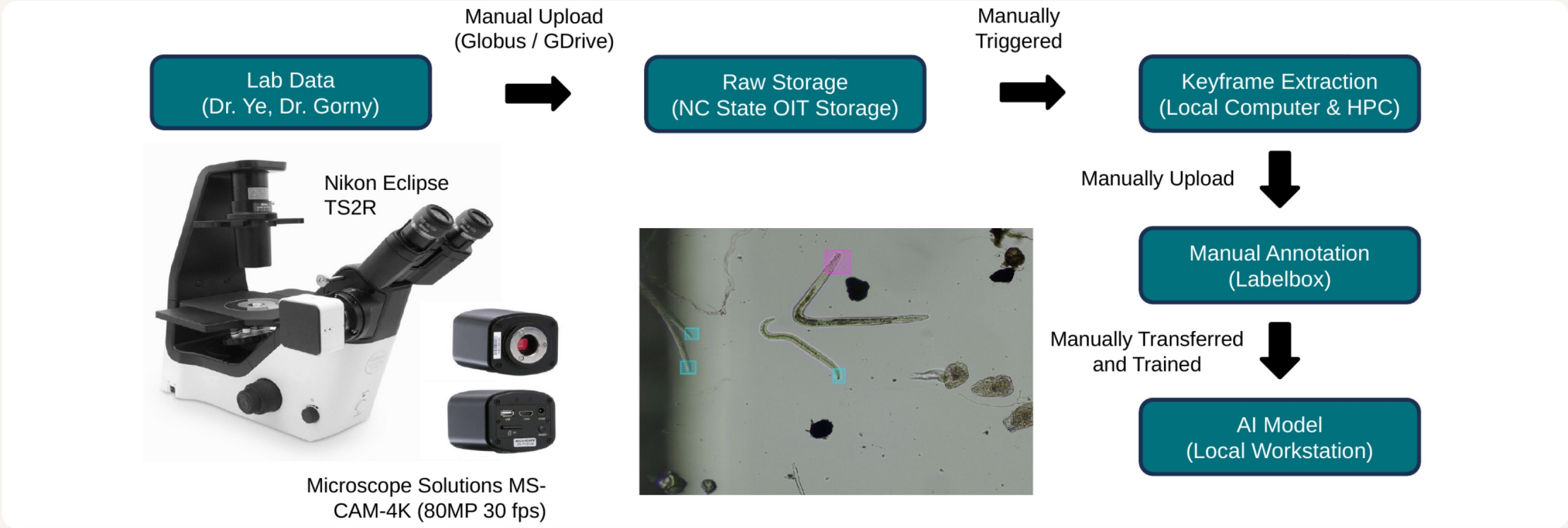
The gap

Real assay samples are full of debris

Jimenez et al., AgriEngineering 7, 356 (2025)

NEMA-AI · PHASE 1

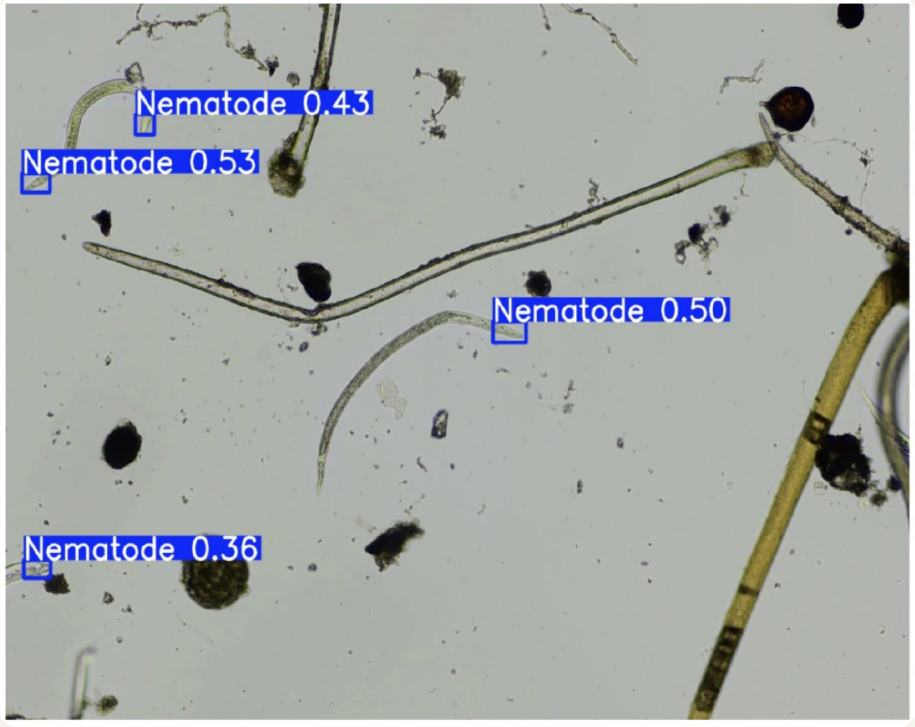
The first data pipeline: many manual handoffs



23 capture sessions at NCDA&CS · video → keyframes → **Labelbox** → model. Every arrow labeled “manually” is a bottleneck.

NEMA-AI · PHASE 2

A first detector, and what its scores tell us



It finds nematodes in debris
Each box comes with a confidence score

But it is unsure
Scores of 0.36–0.53 on clear organisms

Use that uncertainty
Send the least certain images to experts first

WHAT WE LEARNED · NEMA-AI

The bottleneck was annotation, not the model

Detection accuracy (mAP)

Class	mAP
Soybean cyst (SCN cyst)	91.8%
SCN juveniles (J2)	67.9%
Root-knot J2	66.3%
“Other”, hidden head	≥ 21.6%

Where the false detections came from

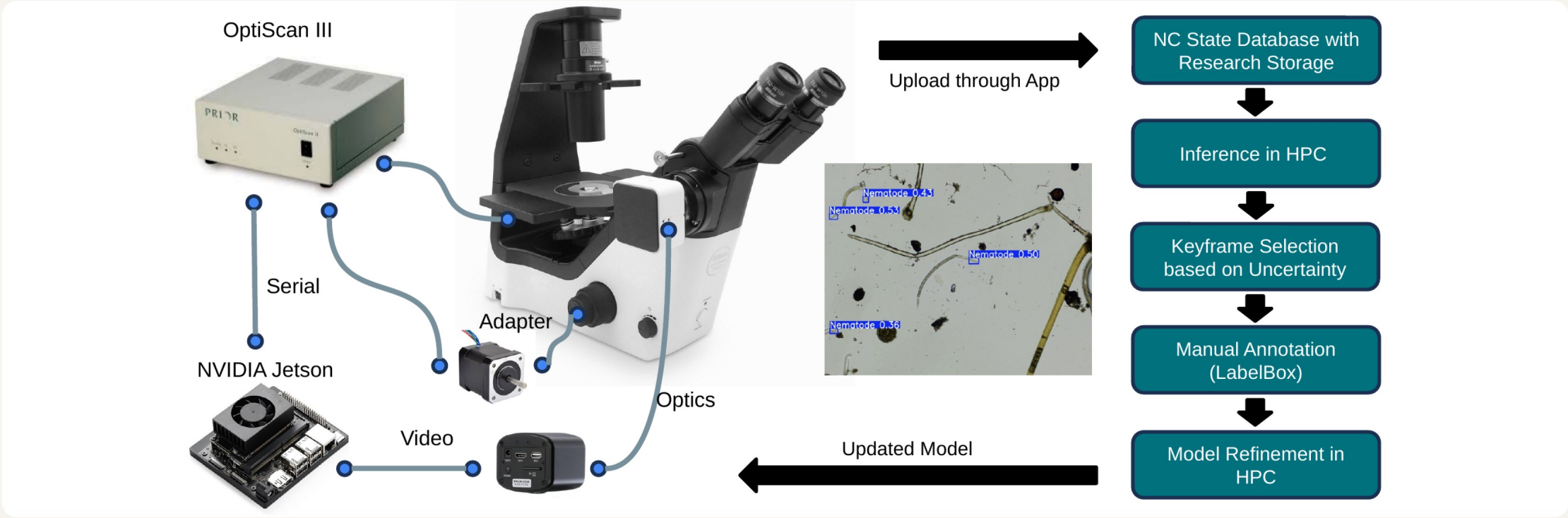
~2/5 Near-misses: need more labels

~1/3 Out-of-focus, never labeled

~1/4 Same organism detected twice

2,700+ annotated images, 6 classes, strongly imbalanced. Two of three fixes are about the labels.

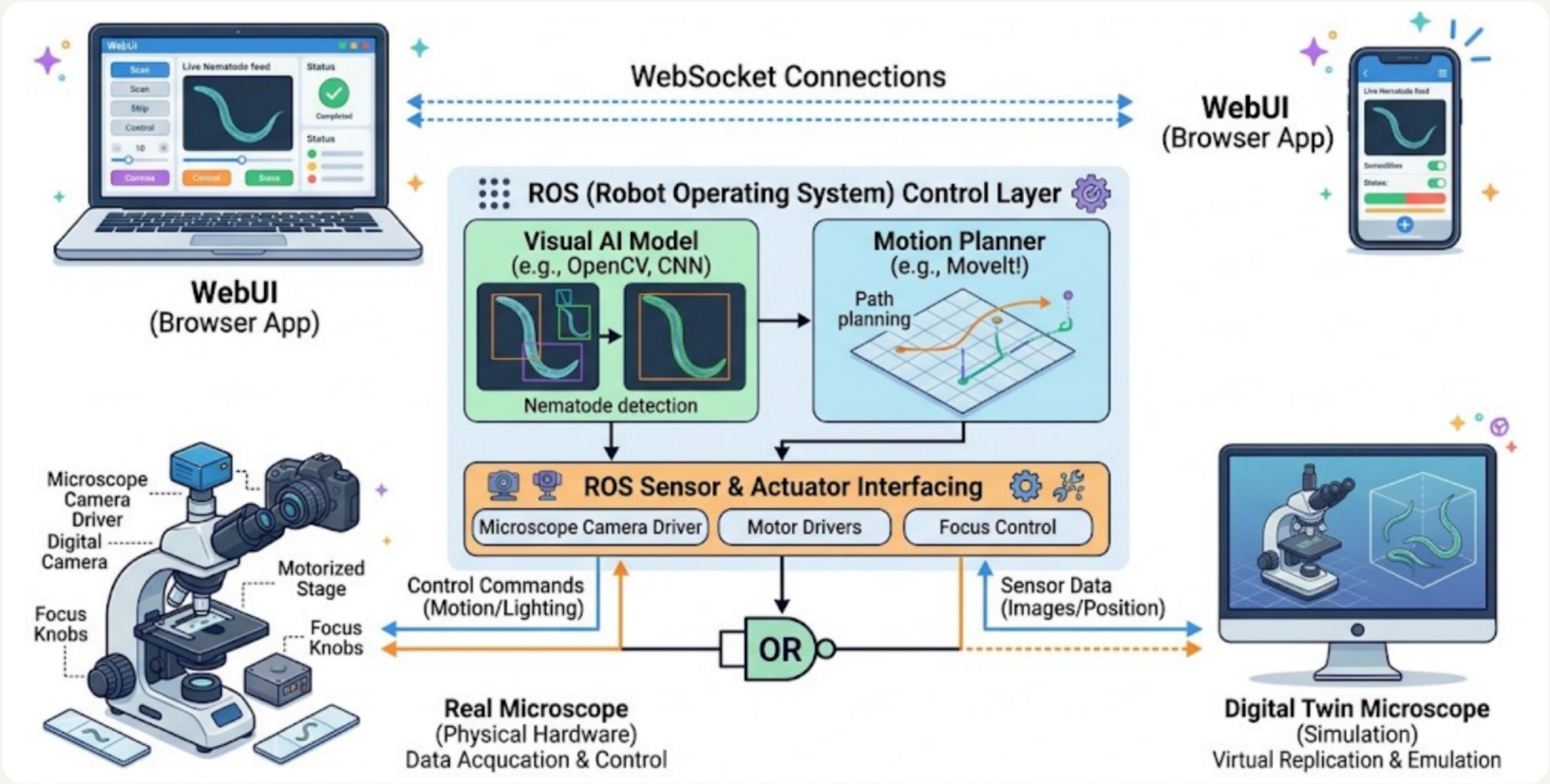
An automated microscope that feeds its own model



Jetson-controlled stage and focus → upload → inference → **pick uncertain frames** → Labelbox → retrain → updated model.

FUTURE WORK · DIGITAL TWIN

Test the microscope in software first



One control layer

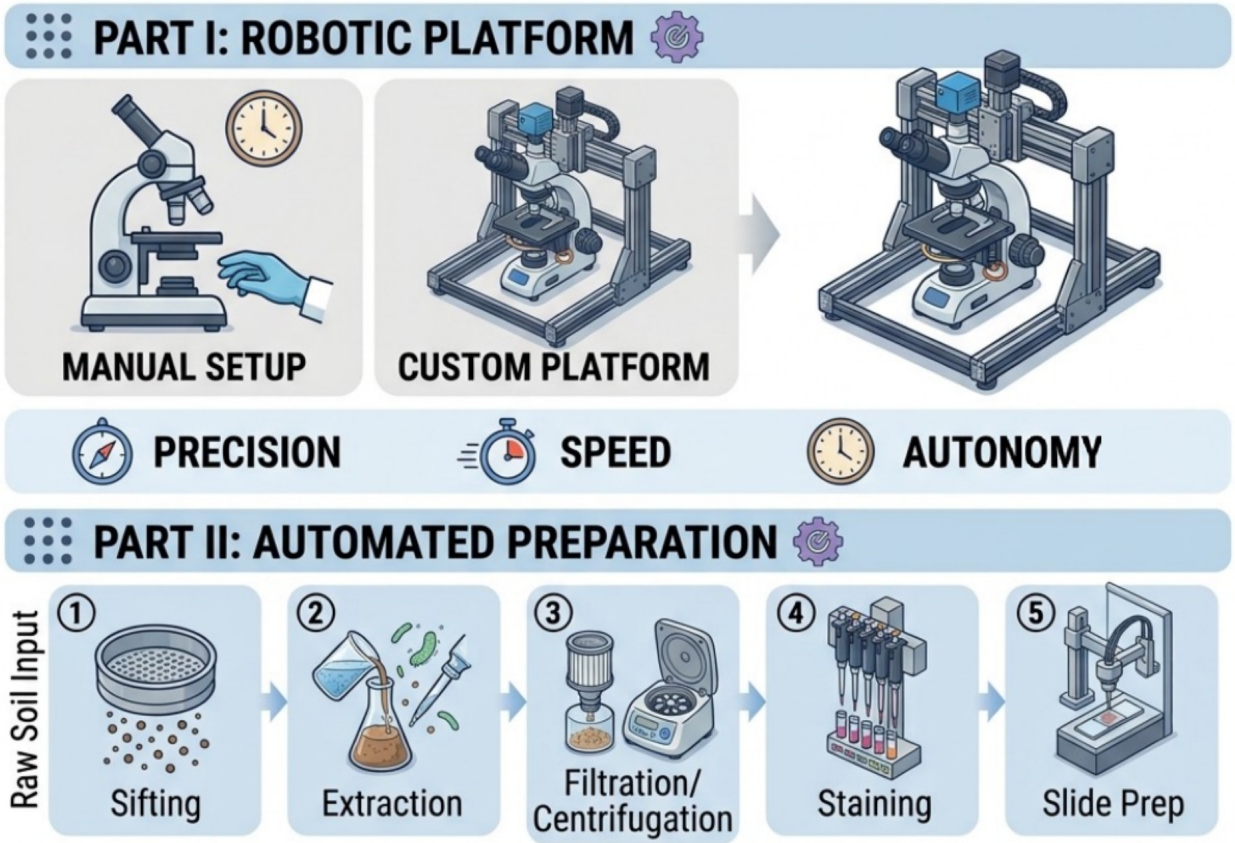
Drives the real or the simulated microscope

Faster development

Scan and focus strategies tested off the lab bench

FUTURE WORK · FULL AUTOMATION

From soil sample to count, with less handling



Imaging platform
Match expert speed; track organisms to avoid double counts

Sample preparation
Less variability, samples in parallel

Goal
Triage ~40% of lab samples

TODAY'S WORKFLOW

From a folder of images to an app, in six steps

1 · Raw
A few hundred frames

→

2 · Pre-label
SAM 2 + prompts

→

3 · Correct
Experts in Labelbox

→

4 · Train
Small detector

→

5 · Expand
Model labels the rest

↓

6 · Deploy
Streamlit app

← The loop: uncertain results from the app go back to Labelbox, and the model is retrained.